

July 2025 CSE 311
CT 4

Name: _____

Student ID: _____

Full Marks: 20

Duration: 25 Minutes

1. Nasir and Rashed are two friends. Nasir wants to send 16-bit data [1010010011100110] to Rashed following block code for product code convention.
 - a) Encode the data to the codeword of (25, 16) block code. [Consider all parity bits as even parity.] [7]
 - b) The codeword received by the channel decoder is [1010001001110010110001100]. Detect and correct the error (if any) following the decoding procedure. [7]
 - c) If the received codeword is [1010001101110010110001100], can the error be corrected? Explain with proper reasons. [6]